



Guião 10– OPSFv3

Objectivo:

- Configurar encaminhamento OSPFv3

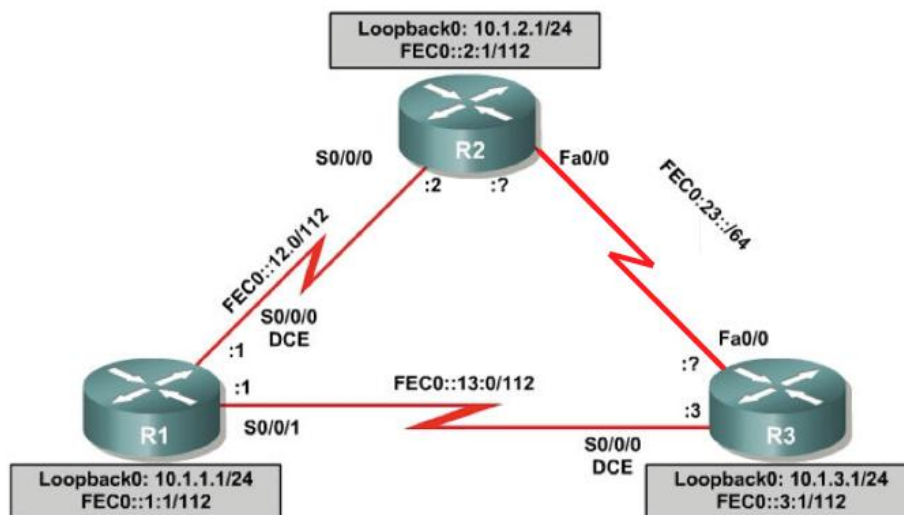


Figure 4 – Topologia da Rede 3

1. Grave o projeto anterior e crie um novo. Desenhe no GNS3 a topologia ilustrada na figura 5. Na configuração dos routers, adicione uma carta NM-4T.

Nota: O link entre o R2 e o R3 é usando FastEthernet. Os links entre R2<->R1 e entre R1<->R3 são usando a interface de série.

2. Configure as interfaces de loopback em cada router com os endereços IPv4 e IPv6 indicados no diagrama. O endereço IPv4 é configurado da maneira normal (“**ip address mascara**”) O endereço IPv6 é semelhante, usando o comando **ipv6 address address/mask**.

```
R1(config)# interface loopback0
R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# ipv6 address FEC0::1:1/112
R2(config)# interface loopback0
R2(config-if)# ip address 10.1.2.1 255.255.255.0
R2(config-if)# ipv6 address FEC0::2:1/112
R3(config)# interface loopback0
R3(config-if)# ip address 10.1.3.1 255.255.255.0
R3(config-if)# ipv6 address FEC0::3:1/112
```

3. Configure os endereços estáticos IPv6

```
R1(config)# interface serial0/0/0
```



```
R1(config-if)# ipv6 address FEC0::12:1/112
R1(config-if)# no keepalive
R1(config-if)# no shutdown
R1(config-if)# interface s0/0/1
R1(config-if)# ipv6 address FEC0::13:1/112
R1(config-if)# no keepalive
R1(config-if)# no shutdown
R2(config)# interface serial0/0/0
R2(config-if)# ipv6 address FEC0::12:2/112
R2 (config-if)# no keepalive
R2(config-if)# no shutdown
R3(config)# interface serial0/0/0
R3(config-if)# ipv6 address FEC0::13:3/112
R3(config-if)# no keepalive
R3(config-if)# no shutdown
```

4. Verifique a conectividade local usando o ping.

5. Execute esta configuração nas interfaces FastEthernet do R2 e R3 tendo em conta as redes do diagrama.

```
R2(config)# interface fastethernet0/0
R2(config-if)# ipv6 address FEC0:23::/64 eui-64
R2(config-if)# no shutdown
R3(config)# interface fastethernet0/0
R3(config-if)# ipv6 address FEC0:23::/64 eui-64
R3(config-if)# no shutdown
```

6. Verifique quais são os endereços IPv6 das interfaces com o comando **show ipv6 interface** or **show ipv6 interface brief**, e faça ping à interface do outro router na mesma rede.

7. Ative o routing IPv6 e o CEF (Cisco Express Forwarding – aceleração de switching L3)

```
R1(config)# ipv6 unicast-routing
R1(config)# ipv6 cef
R2(config)# ipv6 unicast-routing
R2(config)# ipv6 cef
R3(config)# ipv6 unicast-routing
R3(config)# ipv6 cef
```



8. Configure o OSPF IPv6 usando o comando `ipv6 ospf process area <area>` para adicionar uma interface a uma área.

```
R1(config)#interface loopback0
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#interface serial0/0/0
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#interface serial0/0/1
R1(config-if)#ipv6 ospf 1 area 0
R2(config)#interface loopback0
R2(config-if)#ipv6 ospf 1 area 0
R2(config-if)#interface serial0/0/0
R2(config-if)#ipv6 ospf 1 area 0
R2(config-if)#interface fastethernet0/0
R2(config-if)#ipv6 ospf 1 area 0
R3(config)#interface loopback0
R3(config-if)#ipv6 ospf 1 area 0
R3(config-if)#interface serial0/0/0
R3(config-if)#ipv6 ospf 1 area 0
R3(config-if)#interface fastethernet0/0
R3(config-if)#ipv6 ospf 1 area 0
```

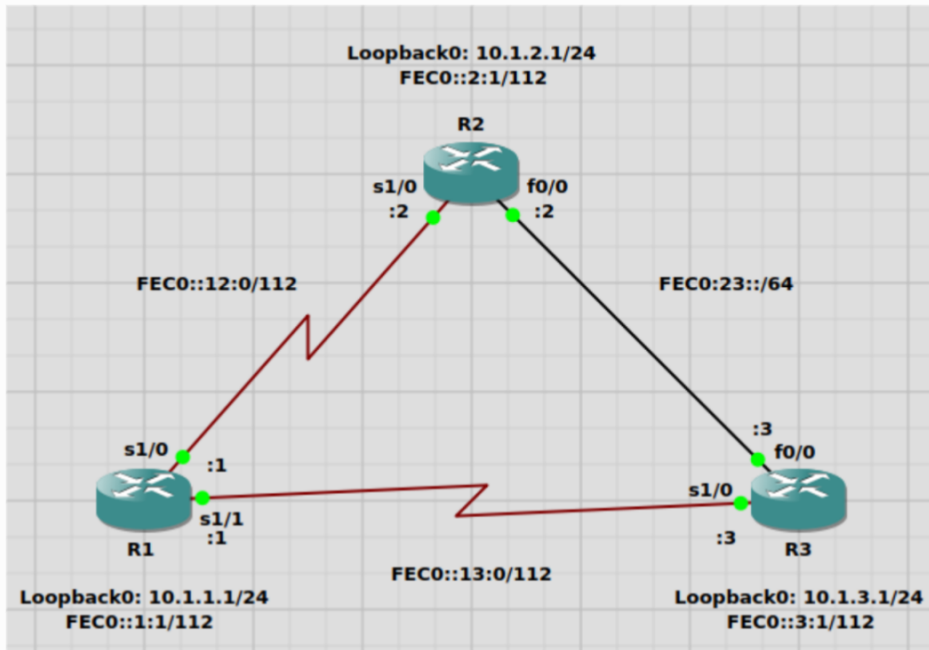
9. Verifique que tem vizinhos OSPFv3 com o comando `show ipv6 ospf neighbor`.

10. Verifique a tabela de encaminhamento de todos os routers `show ipv6 route`.

R1#show ipv6 route **(Registe as respostas para mostrar ao professor)**

OPSFv3

1. Topologia:



2. Configurar interfaces Loopback0 dos routers:

```
R1#conf t
Enter configuration commands, one per line. End
R1(config)#
R1(config)#interface loopback0
R1(config-if)#
*Mar 1 00:06:44.775: %LINEPROTO-5-UPDOWN: Line
state to up
R1(config-if)#ip add
R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#ipv6 add
R1(config-if)#ipv6 address FEC0::1:1/112
% Incomplete command.

R1(config-if)#ipv6 address FEC0::1:1/112
```

```
R2#conf t
Enter configuration commands, one per line. End
R2(config)#interface loopback0
R2(config-if)#int
*Mar 1 00:13:19.663: %LINEPROTO-5-UPDOWN: Line p
state to up
R2(config-if)#ip add
R2(config-if)#ip address 10.1.2.1 255.255.255.0
R2(config-if)#ipv6 address FEC0::2:1/112
```

```
R3#conf t
Enter configuration commands, one per line. End
R3(config)#interface loopback0
R3(config-if)#ip
*Mar  1 00:14:41.779: %LINEPROTO-5-UPDOWN: Line
state to up
R3(config-if)#ip add
R3(config-if)#ip address 10.1.3.1 255.255.255.0
R3(config-if)#ipv6 address FEC0::3:1/112
```

3. Configurar interfaces estáticas (IPv6) dos routers:

```
R1#conf t
Enter configuration commands, one per line
R1(config)#int s1/0
R1(config-if)#ipv6 add
R1(config-if)#ipv6 address FEC0::12:1/112
R1(config-if)#no keep
R1(config-if)#no keepalive
R1(config-if)#no shut
```

```
R1(config-if)#int s1/1
R1(config-if)#ipv6 address FEC0::13:1/112
R1(config-if)#no keepalive
R1(config-if)#no shut
```

```
R2#conf t
Enter configuration commands, one per line
R2(config)#int s1/0
R2(config-if)#ipv6 add
R2(config-if)#ipv6 address FEC0::12:2/112
R2(config-if)#no keep
R2(config-if)#no keepalive
R2(config-if)#no shut
```

```
R3#conf t
Enter configuration commands, one per line
R3(config)#int s1/0
R3(config-if)#ipv6 address FEC0::13:3/112
R3(config-if)#no keep
R3(config-if)#no keepalive
R3(config-if)#no shut
```

4. Existe conectividade local entre os routers:

```
R1#ping FEC0::13:3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0::13:3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/28/32 ms
R1#ping FEC0::12:2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0::12:2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/28/32 ms
```

5. Configuração das interfaces f0/0 dos routers R2 e R3 "automaticamente" com um IPv6 da rede FEC0:23::/64 :

```
R2#conf t
Enter configuration commands, one per line. End with Ctrl-Z.
R2(config)#int f0/0
R2(config-if)#ipv6 add
R2(config-if)#ipv6 address FEC0:23::/64 eui-64
R2(config-if)#no shut
```

```
R3#conf t
Enter configuration commands, one per line. End with Ctrl-Z.
R3(config)#int f0/0
R3(config-if)#ipv6 add
R3(config-if)#ipv6 address FEC0:23::/64 eui-64
R3(config-if)#no shut
```

6.

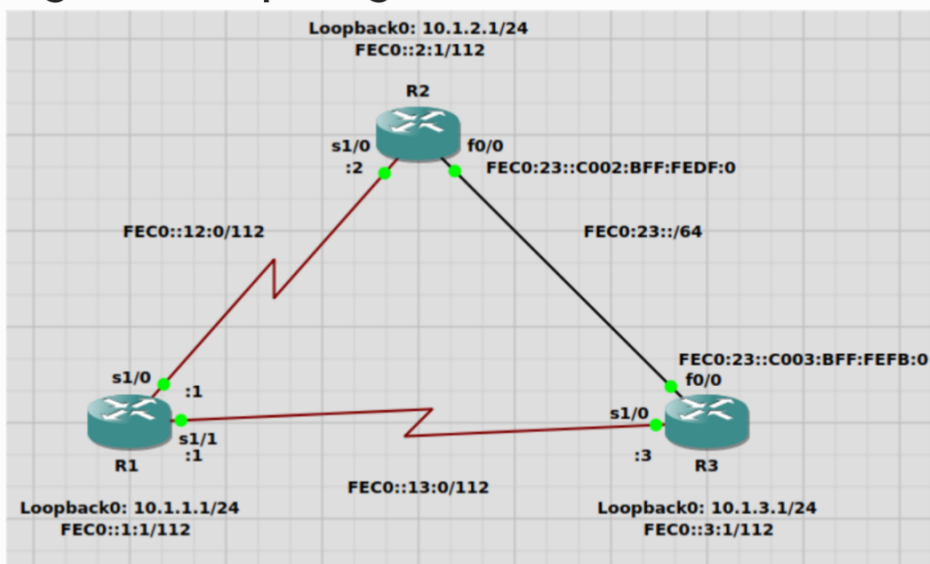
```
R2(config-if)#do sh ipv6 int br
FastEthernet0/0          [up/up]
    FE80::C002:BFF:FEDF:0
    FEC0:23::C002:BFF:FEDF:0
FastEthernet0/1          [administratively down/down]
Serial1/0                [up/up]
    FE80::C002:BFF:FEDF:0
    FEC0::12:2
Serial1/1                [administratively down/down]
Serial1/2                [administratively down/down]
Serial1/3                [administratively down/down]
Loopback0                [up/up]
    FE80::C002:BFF:FEDF:0
    FEC0::2:1
```

```

R3(config-if)#do sh ipv6 int br
FastEthernet0/0                [up/up]
    FE80::C003:BFF:FEFB:0
    FEC0:23::C003:BFF:FEFB:0
FastEthernet0/1                [administratively down/down]
Serial1/0                      [up/up]
    FE80::C003:BFF:FEFB:0
    FEC0::13:3
Serial1/1                      [administratively down/down]
Serial1/2                      [administratively down/down]
Serial1/3                      [administratively down/down]
Loopback0                     [up/up]
    FE80::C003:BFF:FEFB:0
    FEC0::3:1

```

Agora, a topologia ficou:



Fazendo ping à interface f0/0 do R3, como está diretamente conectado ao R2, e ambos já têm IPs da mesma rede, o ping funciona:

```

R2#ping FEC0:23::C003:BFF:FEFB:0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0:23::C003:BFF:FEFB:0, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/45/92 ms

```

7. Ativar o routing IPv6 e o CEF (Cisco Express Forwarding – aceleração de switching L3; otimização do tráfego IPv6):

```

R1(config)#ipv6 unicast-routing
R1(config)#ipv6 cef

```

```

R2(config)#ipv6 unicast-routing
R2(config)#ipv6 cef

```

```
R3#conf t
Enter configuration commands, one
R3(config)#ipv6 unicast-routing
R3(config)#ipv6 cef
```

8. Configurar OSPFv3 nos três routers:

```
R1#conf t
Enter configuration commands, one
R1(config)#interface loopback0
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#int s1/0
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#int s1/1
R1(config-if)#ipv6 ospf 1 area 0
```

```
R2#conf t
Enter configuration commands, one
R2(config)#interface loopback0
R2(config-if)#ipv6 ospf 1 area 0
R2(config-if)#interface s1/0
R2(config-if)#ipv6 ospf 1 area 0
R2(config-if)#interface s1/0
*Mar  1 01:09:21.279: %OSPFv3-5-AD
to FULL, Loading Done
R2(config-if)#interface f0/0
R2(config-if)#ipv6 ospf 1 area 0
```

```
R3#conf t
Enter configuration commands, one
R3(config)#int loopback0
R3(config-if)#ipv6 ospf 1 area 0
R3(config-if)#int s1/0
R3(config-if)#ipv6 ospf 1 area 0
R3(config-if)#int s1/0
*Mar  1 01:13:00.739: %OSPFv3-5-A
to FULL, Loading Done
R3(config-if)#int f0/0
R3(config-if)#ipv6 ospf 1 area 0
```

9. Vizinhos OPSFv3 dos 3 routers (sinalizados através do router ID de cada interface loopback0 de cada neighbor):

```
R1#show ipv6 ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
10.1.3.1	1	FULL/ -	00:00:39	6	Serial1/1
10.1.2.1	1	FULL/ -	00:00:39	6	Serial1/0

```
R2#show ipv6 ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
10.1.3.1	1	FULL/BDR	00:00:31	4	FastEthernet0/0
10.1.1.1	1	FULL/ -	00:00:37	6	Serial1/0

```
R3#show ipv6 ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
10.1.2.1	1	FULL/DR	00:00:35	4	FastEthernet0/0
10.1.1.1	1	FULL/ -	00:00:31	7	Serial1/0

10. Routing Tables dos três routers:

```
R1#sh ipv6 route
```

```
IPv6 Routing Table - 11 entries
```

```
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
```

```
U - Per-user Static route
```

```
I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
```

```
O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
```

```
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
```

```
L FE80::/10 [0/0]
  via ::, Null0
C FEC0::1:0/112 [0/0]
  via ::, Loopback0
L FEC0::1:1/128 [0/0]
  via ::, Loopback0
O FEC0::2:1/128 [110/64]
  via FE80::C002:BFF:FEDF:0, Serial1/0
O FEC0::3:1/128 [110/64]
  via FE80::C003:BFF:FEFB:0, Serial1/1
C FEC0::12:0/112 [0/0]
  via ::, Serial1/0
L FEC0::12:1/128 [0/0]
  via ::, Serial1/0
C FEC0::13:0/112 [0/0]
  via ::, Serial1/1
L FEC0::13:1/128 [0/0]
  via ::, Serial1/1
O FEC0:23::/64 [110/74]
  via FE80::C002:BFF:FEDF:0, Serial1/0
  via FE80::C003:BFF:FEFB:0, Serial1/1
L FF00::/8 [0/0]
  via ::, Null0
```

```
R2#show ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
L   FE80::/10 [0/0]
    via ::, Null0
O   FEC0::1:1/128 [110/64]
    via FE80::C001:BFF:FEC3:0, Serial1/0
C   FEC0::2:0/112 [0/0]
    via ::, Loopback0
L   FEC0::2:1/128 [0/0]
    via ::, Loopback0
O   FEC0::3:1/128 [110/10]
    via FE80::C003:BFF:FEFB:0, FastEthernet0/0
C   FEC0::12:0/112 [0/0]
    via ::, Serial1/0
L   FEC0::12:2/128 [0/0]
    via ::, Serial1/0
O   FEC0::13:0/112 [110/74]
    via FE80::C003:BFF:FEFB:0, FastEthernet0/0
C   FEC0:23::/64 [0/0]
    via ::, FastEthernet0/0
L   FEC0:23::C002:BFF:FEDF:0/128 [0/0]
    via ::, FastEthernet0/0
L   FF00::/8 [0/0]
    via ::, Null0
```

```
R3#sh ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
L   FE80::/10 [0/0]
    via ::, Null0
O   FEC0::1:1/128 [110/64]
    via FE80::C001:BFF:FEC3:0, Serial1/0
O   FEC0::2:1/128 [110/10]
    via FE80::C002:BFF:FEDF:0, FastEthernet0/0
C   FEC0::3:0/112 [0/0]
    via ::, Loopback0
L   FEC0::3:1/128 [0/0]
    via ::, Loopback0
O   FEC0::12:0/112 [110/74]
    via FE80::C002:BFF:FEDF:0, FastEthernet0/0
C   FEC0::13:0/112 [0/0]
    via ::, Serial1/0
L   FEC0::13:3/128 [0/0]
    via ::, Serial1/0
C   FEC0:23::/64 [0/0]
    via ::, FastEthernet0/0
L   FEC0:23::C003:BFF:FEFB:0/128 [0/0]
    via ::, FastEthernet0/0
L   FF00::/8 [0/0]
    via ::, Null0
```